I-Chia Chang (張益嘉)

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EDUCATION

Ph. D. in Mechanical Engineering, Purdue University, West Lafayette, USA *2023 - present*
Thesis Topic: Mobile manipulation of humanoid robots using whole-body contact on moving surfaces

B.S. in Mechanical Engineering, **National Taiwan University**, Taipei, Taiwan *2017 – 2022*
Minor in **Physics**, Overall GPA: **4.13**/4.30 (3.92/4.00), Ranking: **2**/137

PUBLICATIONS [Google Scholar]

I-Chia Chang*, Jacob Stewart*, Petros A. Ioannou, Yan Gu, "Adaptive Rejection of Unknown Hybrid Sinusoidal Disturbances with Application to Humanoid Walking under Unknown Ground Motion," to submit soon. [arXiv] [video] *Equal contributions.

I-Chia Chang, Xinyan Huang, Tzu-Yuan Lin, Sangli Teng, Wenjing Li, Maani Ghaffari, Jingang Yi*, Yan Gu*, "A Survey of Legged Robotics in Non-Inertial Environments: Past, Present, and Future," under review. [arXiv] *Corresponding authors.

Yan Gu*, Guanya Shi*, Fan Shi*, **I-Chia Chang**, Yen-Jen Wang, Qilong Cheng, Zachary Olkin, Ivan Lopez-Sanchez, Yunchu Feng, Jian Zhang, Aaron D. Ames, Hao Su**, and Koushil Sreenath**, ‘‘Evolution of Humanoid Locomotion Control,’’ Science Robotics, 11(117), eaed3973, August 2026 [PDF] [Website] *Equal contributions. **Corresponding authors.

L. C. Y. Tay, **I-Chia Chang**, Y. Gu, “HybridMimic: Hybrid RL-Centroidal Control for Humanoid Motion Mimicking,” in Proc. of IEEE International Conference on Intelligent Robots and Systems (IROS), 2026, accepted. [arXiv] [Code] [Video]

S. R. Iyer, **I-Chia Chang**, Y. Gu, Z. Kingston, “Vectorizing Projection in Manifold-Constrained Motion Planning for Real-Time Whole-Body Control,” in Proc. of IEEE International Conference on Intelligent Robots and Systems (IROS), 2026, accepted. [arXiv] [Project page]

I-Chia Chang and Pei-Chun Lin. ‘‘Energy-Efficient and Stride-Level Hybrid Locomotion of a Leg-Wheel Transformable Robot.’’ ASME. J. Mechanisms Robotics 2025. [Link] [PDF] [Video]

Jacob Stewart*, **I-Chia Chang***, Yan Gu, Petros A. Ioannou. “Adaptive Ankle Torque Control for Bipedal Humanoid Walking on Surfaces with Unknown Horizontal and Vertical Motion,” 2025 American Control Conference (ACC), Denver, CO, USA, 2025. [arXiv] *Equal contributions.

Kuan-Lun Lu, **I-Chia Chang**, Wei-Shun Yu, Pei-Chun Lin. “Trajectory Optimization Strategy That Considers Body Tip-Over Stability, Limb Dynamics, and Motion Continuity in Legged Robots,” IEEE International Conference on Robotics and Automation (ICRA) 2024. [PDF]

Pei-Chun Huang, **I-Chia Chang**, Wei-Shun Yu, Pei-Chun Lin. “Body Velocity Estimation in a Leg–Wheel Transformable Robot without A Priori Knowledge of Leg–Wheel Ground Contacts,” IEEE International Conference on Robotics and Automation (ICRA) 2024. [PDF]

I-Chia Chang and Pei-Chun Lin. “Dynamic turning and running of a hexapod robot using a separated and laterally arranged two-leg model.” Bioinspiration & Biomimetics 2023. [Link] [Video]

Chun-Kai Huang, Chia-Jui Hu, **I-Chia Chang**, Pei-Chun Lin, “A passive planar model with rolling contact and its application for bounding and pronking on a quadruped robot,” Journal of Mechanics, vol. 38, pp621-642, 2022 [Link]

I-Chia Chang, Chih-Hsiang Hsu, Hong-Sheng Wu, and Pei-Chun Lin. “An analysis of the rolling dynamics of a hexapod robot using a three-dimensional rolling template,” Nonlinear Dynamics, vol. 109, pp631-655, 2022 [Link] [PDF]

SCHOLARSHIP AND HONORS RECEIVED

Best Overall Poster, 2026 ICON Student Research Conference
Ross Fellowship, Purdue University, 2023
The 2021 Bachelor's Thesis Award, 1st Place, National Taiwan University, 2021
College Student Research Scholarship, Ministry of Science and Technology, Taiwan, 2020

ACADEMIC SERVICES

Workshop organizer:

- Perception and Decision Making for Athletic Humanoid Robotics workshop at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026. [Web]

Journal Reviewer:

- IEEE Robotics and Automation Letters (RA-L), IEEE/ASME Transactions on Mechatronics (T-Mech), IEEE Transactions on Automation Science and Engineering (T-ASE)

Conference Reviewer:

- IEEE International Conference on Robotics and Automation (ICRA), American Control Conference (ACC), IEEE-RAS International Conference on Humanoid Robots (Humanoid), International Conference on Biomedical Robotics and Biomechatronics (BioRob)

Grant Reviewer:

- Purdue Graduate Student Government (PGSG) grant reviewer

TEACHING EXPERIENCE

Mentor for undergraduate and master students in TRACE Lab, Purdue *2023 – Present*

- Mentored more than 6 undergraduate and master students to conduct research about the data-driven state estimation, humanoid control using model-based and/or learning methods, and sensor design for human-human and human-robot interaction.

Mentor for undergraduate students and summer interns in Bio-inspired Robotics Lab *2021 – 2023*

- Mentored 3 undergraduate students to conduct research about the energy efficiency optimization for a dynamic running robot with experimental and simulation methods
- Mentored 2 high school students to build a snake robot with actuated universal joints, and a six-wheel drive rover with a self-balancing platform to overcome broken natural terrain

Teaching Assistant for the course "Designing Your NTU," National Taiwan University *2021 – 2022*

- Led the discussions about imagination and implementation of education in future universities
- Proposed modifications to the curriculum design to improve the course evaluation scores by 10%

Teacher Education Program, National Taiwan University *2019 – 2022*

- Took thirteen courses to learn and practice the principles of learning and teaching

SERVICE AND LEADERSHIP EXPERIENCE

Founder of the NTU for FIRST Robotics Competition (FRC) club *2018 – 2023*

- Established a sustainable club to voluntarily provide high schoolers a high-quality STEM educational support, including workshops, lectures, and mentoring
- Systematized a mentor training process to enable club members to support high school students
- Supported more than four high schools by providing self-trained mentors, results of robotics research and prototyping, and the advanced engineering knowledge learned from college

Mentor lead of FRC team 8180 *2019 – 2020*

- Led a group of 8 mentors to voluntarily provide students who had no or little access to STEM education with an inspiring robotics teamwork experience
- Taught members the kinematics, dynamics, mechatronics, and metal machining skills required
- Awarded for the winner in the "2019 Taiwan FRC Off-Season Forum in CTSP."

SKILLS

Programming Language: C/C++, Python, MATLAB, LabVIEW, Simulink, Webots.
Mechanical Design Software: AutoCAD, SolidWorks, Inventor